ENGINEERING ECONOMICS

CE 655 | Year III / Part II

Chapter 7: Depreciation and Income Tax

Questions & Answers — TU Past Exam Papers (2070–2082)

Tribhuvan University | Institute of Engineering

Chapter 7 Syllabus Overview (5 Hours)

Topics:

a) Introduction to Depreciation — Definition, Causes, Purpose, Terminology
b) Depreciation Methods:
    - Straight Line (SL)
    - Declining Balance (DB)
    - Double Declining Balance (DDB)
    - Sum-of-Years-Digits (SYD)
    - Units of Production (UOP)
c) Comparison of Methods
d) Income Tax — Taxable Income, Tax Rate, After-Tax Cash Flow
e) After-Tax Economic Analysis — AT-NPW, AT-AW, AT-IRR
f) Effect of Depreciation Method on After-Tax NPW

Section A: Depreciation — Concepts and Terminology

Q1. Define depreciation. Explain its causes, purposes, and key terminology. [2082 Kartik Q9a, 2081 Ashwin Q9a, 2079 Jestha Q9a — 4 Marks]

Answer:

Definition

Depreciation is the systematic allocation of a capital asset's cost over its useful life, representing the gradual decline in value due to use, wear, obsolescence, or the passage of time. In engineering economics it is critical because it creates a tax-deductible charge that reduces income tax — a real cash benefit even though depreciation itself involves no cash outflow.

Causes of Depreciation

1. Physical Deterioration: Moving parts wear, surfaces corrode, and structures fatigue through normal use and ageing.
2. Functional Obsolescence: The asset becomes inadequate for current needs due to changes in production requirements, even while physically operational.
3. Technological Obsolescence: Newer, more efficient technology makes the existing asset economically inferior. Example: analogue instruments replaced by digital systems.
4. Depletion: For natural resources (mines, wells, timber), the physical quantity consumed progressively reduces remaining value.
5. Accident or Casualty: Unexpected damage or destruction reduces asset value or shortens remaining useful life.

Purposes of Depreciation

1. Tax Deduction: Depreciation is deductible from gross income, reducing taxable income and thereby lowering annual tax payments — a direct cash-flow benefit.
2. Capital Recovery: Distributes the capital investment cost over the years the asset generates revenue, ensuring true profit is only recognised after full cost recovery.
3. Asset Valuation: Maintains realistic book values on the balance sheet.

Key Depreciation Terminology

Term	Symbol	Definition
First Cost (Unadjusted Basis)	B	Original purchase price plus all costs to place asset in service
Salvage Value	S	Estimated market value at end of depreciable life
Book Value at End of Year k	BV_k	B minus accumulated depreciation: $BV_k = B - \Sigma d_j$

Term	Symbol	Definition
Depreciable Life	n	Years over which the asset is depreciated
Annual Depreciation	d_k	Depreciation amount allocated to year k
Accumulated Depreciation	ΣD_k	Sum of all charges from year 1 through year k

Q2. Derive the Straight Line (SL) depreciation formula. B = Rs. 1,20,000; S = Rs. 20,000; n = 8 years. Prepare the complete SL schedule. [2081 Chaitra Q9a, 2079 Ashwin Q9a, 2074 Bhadra Q9a — 5 Marks]

Answer:

SL Method — Derivation

The Straight Line method allocates an equal depreciation charge to each year, assuming the asset loses value at a constant rate. Total depreciable amount = B − S, spread equally over n years:

```
Annual Depreciation:   d_SL = (B − S) / n   [constant for all years]
Book Value at end of year k:   BV_k = B − k × d_SL
```

B = Rs. 1,20,000; S = Rs. 20,000; n = 8 years

```
d_SL = (1,20,000 − 20,000) / 8 = Rs. 12,500 per year
```

Year (k)	d_k (Rs.)	Accumulated Depr. (Rs.)	BV_k (Rs.)
0	—	—	1,20,000
1	12,500	12,500	1,07,500
2	12,500	25,000	95,000
3	12,500	37,500	82,500
4	12,500	50,000	70,000
5	12,500	62,500	57,500
6	12,500	75,000	45,000
7	12,500	87,500	32,500
8	12,500	1,00,000	20,000 = S ✓

Verification: Total = 8 × 12,500 = Rs. 1,00,000 = B−S ✓. Final BV = Rs. 20,000 = S ✓.

Q3. Explain DB and DDB methods. B = Rs. 80,000; S = Rs. 8,000; n = 5 years. Prepare the DDB schedule and compare with SL. [2082 Kartik Q9b, 2080 Ashwin Q9a, 2078 Baishakh Q8 — 6 Marks]

Answer:

Declining Balance (DB) Method

The DB method applies a fixed rate to the current book value each year, producing decreasing charges — accelerated depreciation. The rate is a multiple m of the straight-line rate:

```
DB Rate: d_rate = m/n    (m=1.5 for 150%-DB; m=2 for DDB)
d_k = d_rate × BV_{k-1};    BV_k = B × (1 − d_rate)^k
```

BV must never fall below S. If the formula would push BV < S, limit d_k = BV_{k-1} − S, then stop depreciating.

DDB (Double Declining Balance) — m = 2

```
DDB rate = 2/n = 2/5 = 40%
SL: d = (80,000−8,000)/5 = Rs. 14,400/year for comparison
```

Year k	BV_{k-1} (Rs.)	d_DDB = 0.40 × BV (Rs.)	Limit (BV ≥ 8,000)	BV_k (Rs.)
1	80,000	32,000	48,000 > 8,000 ✓	48,000
2	48,000	19,200	28,800 > 8,000 ✓	28,800
3	28,800	11,520	17,280 > 8,000 ✓	17,280
4	17,280	6,912	10,368 > 8,000 ✓	10,368
5	10,368	2,368 (limited)	10,368−2,368=8,000 = S ✓	8,000

Year 5: DDB gives 0.40×10,368 = 4,147 → BV = 6,221 < S(8,000). Limit: d_5 = 10,368 − 8,000 = 2,368.

SL vs. DDB Comparison

Year	d_SL (Rs.)	BV_SL (Rs.)	d_DDB (Rs.)	BV_DDB (Rs.)
1	14,400	65,600	32,000	48,000
2	14,400	51,200	19,200	28,800
3	14,400	36,800	11,520	17,280
4	14,400	22,400	6,912	10,368
5	14,400	8,000	2,368	8,000
Total	72,000	—	72,000	—

Both depreciate the same total (Rs. 72,000). DDB front-loads charges, giving larger tax deductions in years 1–3, producing a higher present value of tax savings.

Q4. Explain the SYD method. B = Rs. 1,00,000; S = Rs. 10,000; n = 6 years. Prepare the complete SYD schedule. [2081 Ashwin Q9b, 2079 Chaitra Q9a, 2077 Chaitra Q9a — 5 Marks]

Answer:

Sum-of-Years-Digits (SYD) — Concept

SYD is an accelerated method assigning a decreasing fraction of the depreciable amount (B−S) to each year. The fraction numerator is the remaining life; denominator is the sum of all year-digits.

$$SOYD = n(n+1)/2$$

$$d_k = [(n-k+1) / SOYD] \times (B-S)$$

B = Rs. 1,00,000; S = Rs. 10,000; n = 6 years; Depreciable = Rs. 90,000

$$SOYD = 6\times7/2 = 21$$

Year k	Remaining Life	Fraction	d_k (Rs.)	Acc. Depr. (Rs.)	BV_k (Rs.)
0	—	—	—	—	1,00,000
1	6	6/21	25,714	25,714	74,286
2	5	5/21	21,429	47,143	52,857
3	4	4/21	17,143	64,286	35,714
4	3	3/21	12,857	77,143	22,857
5	2	2/21	8,571	85,714	14,286
6	1	1/21	4,286	90,000	10,000 = S ✓
Total	21	21/21	90,000		

Verification: Total = Rs. 90,000 = B−S ✓. Final BV = Rs. 10,000 = S ✓.

Q5. Compare SL, SYD, and DDB for B = Rs. 50,000; S = Rs. 5,000; n = 5 years. Comment on tax effects. [2080 Chaitra Q9a, 2081 Chaitra Q9b — 6 Marks]

Answer:

B = Rs. 50,000; S = Rs. 5,000; n = 5 years; Depreciable = Rs. 45,000

SL: d = 45,000/5 = Rs. 9,000/year

DDB: rate = 2/5 = 40%; SYD: SOYD = 15

Year	d_SL (Rs.)	BV_SL (Rs.)	d_SYD (Rs.)	BV_SYD (Rs.)	d_DDB (Rs.)	BV_DDB (Rs.)
1	9,000	41,000	15,000	35,000	20,000	30,000
2	9,000	32,000	12,000	23,000	12,000	18,000
3	9,000	23,000	9,000	14,000	7,200	10,800
4	9,000	14,000	6,000	8,000	4,320	6,480
5	9,000	5,000	3,000	5,000	1,480*	5,000
Total	45,000	—	45,000	—	45,000	—

**DDB Year 5 limited: 0.40×6,480=2,592 → BV=3,888<5,000. Limit d_5=6,480−5,000=1,480.*

Tax Effect Analysis

- All three methods produce identical total depreciation (Rs. 45,000) and thus identical total lifetime taxes.
- However, DDB provides the largest deductions in early years (Rs. 20,000 in Year 1 vs. Rs. 9,000 for SL), reducing taxes most in early years.
- SYD is intermediate: Year 1 deduction = Rs. 15,000 vs. Rs. 9,000 (SL) and Rs. 20,000 (DDB).

Ranking by tax advantage (from best to worst PV of tax savings): DDB > SYD > SL. Accelerated methods are preferred where tax regulations permit, as they defer taxes and improve the present value of after-tax cash flows.

Q6. B = Rs. 90,000; S = Rs. 6,000; n = 7 years. Prepare depreciation schedules for (a) SL, (b) SYD, (c) DDB. [2079 Jestha Q9b, 2076 Baisakh Q9a — 6 Marks]

Answer:

B = Rs. 90,000; S = Rs. 6,000; n = 7 years; Depreciable = Rs. 84,000

SL: d = 84,000/7 = Rs. 12,000/year

DDB rate: 2/7 = 28.57%; SOYD = 7×8/2 = 28

(a) Straight Line

Year	d_SL (Rs.)	BV_k (Rs.)
1	12,000	78,000

Year	d_SL (Rs.)	BV_k (Rs.)
2	12,000	66,000
3	12,000	54,000
4	12,000	42,000
5	12,000	30,000
6	12,000	18,000
7	12,000	6,000 = S ✓

(b) Sum-of-Years-Digits

Year k	Fraction	d_SYD (Rs.)	BV_k (Rs.)
1	7/28	21,000	69,000
2	6/28	18,000	51,000
3	5/28	15,000	36,000
4	4/28	12,000	24,000
5	3/28	9,000	15,000
6	2/28	6,000	9,000
7	1/28	3,000	6,000 = S ✓

(c) DDB at 28.57%

Year k	BV_{k-1} (Rs.)	d_k (Rs.)	BV_k (Rs.)
1	90,000	25,713	64,287
2	64,287	18,367	45,920
3	45,920	13,119	32,801
4	32,801	9,372	23,429
5	23,429	6,694	16,735
6	16,735	4,782	11,953
7	11,953	5,953 (limited)	6,000 = S ✓

Year 7: 0.2857×11,953=3,415 → BV=8,538 > S(6,000). Years 6,7 must be re-checked. After year 6: BV=11,953. DDB year 7: d=3,415, BV=8,538. Not at S yet. This means the final year's depreciation must be set manually: d_7 = 11,953−6,000 = 5,953 (since we must reach S by end of life).

Section B: Units of Production Method

Q7. Explain Units of Production (UOP) depreciation. B = Rs. 2,40,000; S = Rs. 15,000; total life production = 1,50,000 units. Annual production: 28,000; 32,000; 35,000; 30,000; 25,000 units. Find annual depreciation. [2079 Ashwin Q9b, 2074 Bhadra Q9a — 4 Marks]

Answer:

UOP Method — Concept

The UOP method allocates depreciation in proportion to actual usage — measured in units produced, hours operated, or kilometres driven. It is appropriate when wear correlates more with use than with time.

```
Depreciation per unit:   d_unit = (B − S) / Total estimated lifetime production
Annual depreciation:   d_k = d_unit × (Actual units produced in year k)
```

B = Rs. 2,40,000; S = Rs. 15,000; Total units = 1,50,000

```
d_unit = (2,40,000 − 15,000) / 1,50,000 = Rs. 1.50 per unit
```

Year	Units Produced	d_k = 1.50 × Units (Rs.)	Acc. Depr. (Rs.)	BV_k (Rs.)
0	—	—	—	2,40,000
1	28,000	42,000	42,000	1,98,000
2	32,000	48,000	90,000	1,50,000
3	35,000	52,500	1,42,500	97,500
4	30,000	45,000	1,87,500	52,500
5	25,000	37,500	2,25,000	15,000 = S ✓

Total units = 1,50,000 ✓. Total depreciation = Rs. 2,25,000 = B−S ✓.

Section C: Income Tax and After-Tax Cash Flow Analysis

Q8. Explain income tax concepts. Define GI, OE, BTCF, Depreciation, TI, Tax, NATI, and ATCF. [2082 Kartik Q10a, 2081 Chaitra Q9c, 2080 Ashwin Q9b — 4 Marks]

Answer:

Relevance of Income Tax

Income taxes are a significant cash outflow that must be included for accurate project evaluation. An after-tax analysis provides the true economic picture. The depreciation method chosen directly affects taxable income and thus the timing of tax payments, making it an important engineering economic decision.

Term	Formula / Definition
Gross Income (GI)	Total revenues from all sources
Operating Expenses (OE)	All cash operating costs (wages, materials, energy). Excludes depreciation and interest.
Before-Tax Cash Flow (BTCF)	BTCF = GI − OE (actual cash generated, before tax)
Depreciation (D_k)	Annual non-cash depreciation charge — deductible for tax, but not a cash outflow
Taxable Income (TI)	TI = GI − OE − D_k = BTCF − D_k
Income Tax (T)	T = t × TI where t = corporate tax rate
Net After-Tax Income (NATI)	NATI = TI − T = TI(1−t)
After-Tax Cash Flow (ATCF)	ATCF = BTCF − T = NATI + D_k (key figure for economic analysis)

Tax shield from depreciation in year k = t × d_k (real annual cash saving)

Accelerated depreciation creates larger tax shields in early years, improving the NPW of after-tax cash flows through the time value of money — even though the total lifetime tax paid is the same.

Q9. B = Rs. 1,50,000; S = 0; n = 5 years; Annual Revenue = Rs. 70,000; Annual O&M = Rs. 20,000; t = 35%; After-tax MARR = 10%. SL depreciation. Find ATCF schedule and AT-NPW. [2082 Kartik Q10b, 2081 Ashwin Q10a, 2080 Chaitra Q10 — 8 Marks]

Answer:

B = Rs. 1,50,000; S = 0; n = 5 yrs; GI = Rs. 70,000/yr; OE = Rs. 20,000/yr; t = 35%; MARR = 10%

BTCF = 70,000 − 20,000 = Rs. 50,000/year

d_SL = 1,50,000/5 = Rs. 30,000/year

Annual ATCF (Uniform, Years 1–5)

Item	Calculation	Amount (Rs.)
BTCF	GI − OE = 70,000 − 20,000	50,000
Less: Depreciation (d_SL)	Non-cash deduction	30,000
Taxable Income (TI)	BTCF − d = 50,000 − 30,000	20,000
Income Tax (T)	0.35 × 20,000	7,000
ATCF	BTCF − T = 50,000 − 7,000	43,000

Year	Invest. (Rs.)	BTCF (Rs.)	d (Rs.)	TI (Rs.)	Tax (Rs.)	ATCF (Rs.)
0	−1,50,000	—	—	—	—	−1,50,000
1–5		50,000	30,000	20,000	7,000	43,000

After-Tax NPW

$$(P/A,\ 10\%,\ 5) = 3.7908$$

$$\text{AT-NPW} = -1{,}50{,}000 + 43{,}000 \times 3.7908 = -1{,}50{,}000 + 1{,}63{,}004 = \text{Rs. } 13{,}004$$

AT-NPW = Rs. 13,004 > 0. Project is acceptable at 10% after-tax MARR.

Before-tax NPW = −1,50,000 + 50,000(3.7908) = Rs. 39,540. Taxes reduce NPW by Rs. 26,536, highlighting the importance of after-tax analysis.

Q10. Repeat Q9 using DDB depreciation. Compare AT-NPW with SL and explain the difference. [2081 Chaitra Q10, 2079 Ashwin Q10a, 2079 Jestha Q10 — 8 Marks]

Answer:

Same data: B = Rs. 1,50,000; S = 0; n = 5 yrs; BTCF = Rs. 50,000/yr; t = 35%; MARR = 10%

$$\text{DDB rate} = 2/5 = 40\%$$

Year	BV_{k-1} (Rs.)	d_DDB (Rs.)	BV_k (Rs.)
1	1,50,000	60,000	90,000
2	90,000	36,000	54,000
3	54,000	21,600	32,400
4	32,400	12,960	19,440

Year	BV_{k-1} (Rs.)	d_DDB (Rs.)	BV_k (Rs.)
5	19,440	19,440	0 = S ✓

Year	BTCF (Rs.)	d_DDB (Rs.)	TI (Rs.)	Tax @35% (Rs.)	ATCF (Rs.)
1	50,000	60,000	−10,000	−3,500 (credit)	53,500
2	50,000	36,000	14,000	4,900	45,100
3	50,000	21,600	28,400	9,940	40,060
4	50,000	12,960	37,040	12,964	37,036
5	50,000	19,440	30,560	10,696	39,304

Year 1: d (60,000) > BTCF (50,000) → tax loss of −10,000 → tax credit of Rs. 3,500 → ATCF = 53,500.

```
(P/F,10%): yr1=0.9091, yr2=0.8264, yr3=0.7513, yr4=0.6830, yr5=0.6209
```

```
AT-NPW_DDB = -1,50,000 + 53,500(0.9091)+45,100(0.8264)+40,060(0.7513)+37,036(0.6830)+39,304(0.6209)
= -1,50,000 + 48,637+37,271+30,097+25,296+24,407 = Rs. 15,708
```

Method	Total Tax (Rs.)	AT-NPW (Rs.)	Difference
SL	35,000	13,004	Base
DDB	35,000	15,708	+Rs. 2,704 (better)

DDB yields Rs. 2,704 higher AT-NPW. Both pay exactly Rs. 35,000 total tax, but DDB defers taxes to later years. Earlier tax savings are worth more due to time value of money.

Q11. B = Rs. 2,00,000; S = Rs. 20,000; n = 8 years. Revenue = Rs. 90,000/yr; O&M = Rs. 25,000/yr; t = 40%; MARR = 12%. SL depreciation. Find (a) ATCF, (b) AT-NPW, (c) AT-AW. [2080 Chaitra Q10b, 2078 Chaitra Q10 — 8 Marks]

Answer:

B = Rs. 2,00,000; S = Rs. 20,000; n = 8 yrs; BTCF = Rs. 65,000/yr; t = 40%; MARR = 12%

```
d_SL = (2,00,000 - 20,000)/8 = Rs. 22,500/year
```

(a) Annual ATCF

```
TI = 65,000 - 22,500 = Rs. 42,500;   Tax = 0.40 × 42,500 = Rs. 17,000
```

ATCF (yr 1–7) = 65,000 – 17,000 = Rs. 48,000

ATCF (yr 8) = 48,000 + 20,000 (salvage, BV=S so no gain tax) = Rs. 68,000

Year	Invest. (Rs.)	BTCF (Rs.)	d (Rs.)	TI (Rs.)	Tax (Rs.)	ATCF (Rs.)
0	−2,00,000	—	—	—	—	−2,00,000
1–7		65,000	22,500	42,500	17,000	48,000
8		65,000+20,000	22,500	42,500	17,000	68,000

(b) AT-NPW at 12%

(P/A,12%,7) = 4.5638; (P/F,12%,8) = 0.4039

AT-NPW = −2,00,000 + 48,000(4.5638) + 68,000(0.4039)

= −2,00,000 + 2,19,062 + 27,465 = Rs. 46,527

(c) AT-AW

AT-AW = 46,527 × (A/P,12%,8) = 46,527 × 0.20130 = Rs. 9,366/year

AT-NPW = Rs. 46,527 > 0 and AT-AW = Rs. 9,366 > 0. Accept the investment.

Q12. B = Rs. 1,00,000; S = Rs. 10,000; n = 5 yrs; t = 30%; MARR = 10%. SL depreciation. Find ATCF and AT-NPW. [2082 Kartik Q10c, 2078 Baishakh Q9b, 2076 Baisakh Q9 — 6 Marks]

Answer:

Year	Revenue (Rs.)	O&M (Rs.)	BTCF (Rs.)
1	60,000	15,000	45,000
2	70,000	17,000	53,000
3	75,000	18,000	57,000
4	65,000	20,000	45,000
5	55,000	22,000	33,000

d_SL = (1,00,000 – 10,000)/5 = Rs. 18,000/year

Year	BTCF (Rs.)	d (Rs.)	TI (Rs.)	Tax @30% (Rs.)	ATCF (Rs.)
1	45,000	18,000	27,000	8,100	36,900
2	53,000	18,000	35,000	10,500	42,500
3	57,000	18,000	39,000	11,700	45,300

Year	BTCF (Rs.)	d (Rs.)	TI (Rs.)	Tax @30% (Rs.)	ATCF (Rs.)
4	45,000	18,000	27,000	8,100	36,900
5	33,000	18,000	15,000	4,500	28,500

Year 5 ATCF includes salvage of Rs. 10,000 (BV = S, no gain tax): Total ATCF(5) = 28,500 + 10,000 = Rs. 38,500.

$$(P/F,10\%):\ yr1=0.9091,\ yr2=0.8264,\ yr3=0.7513,\ yr4=0.6830,\ yr5=0.6209$$

$$AT\text{-}NPW = -1,00,000 + 36,900(0.9091)+42,500(0.8264)+45,300(0.7513)+36,900(0.6830)+38,500(0.6209)$$

$$= -1,00,000 + 33,546+35,122+34,034+25,203+23,905 = Rs.\ 51,810$$

AT-NPW = Rs. 51,810 > 0. Accept.

Section D: After-Tax IRR and SYD After-Tax Analysis

Q13. Find after-tax IRR. B = Rs. 80,000; S = 0; n = 5 years; BTCF = Rs. 35,000/yr; t = 40%; SL depreciation. [2080 Ashwin Q10b, 2078 Chaitra Q10 — 6 Marks]

Answer:

$$d_SL = 80,000/5 = Rs.\ 16,000/year$$

$$TI = 35,000 - 16,000 = Rs.\ 19,000; \quad Tax = 0.40 \times 19,000 = Rs.\ 7,600$$

$$ATCF = 35,000 - 7,600 = Rs.\ 27,400/year\ (uniform,\ years\ 1\text{-}5)$$

Solving for AT-IRR (AT-NPW = 0)

$$0 = -80,000 + 27,400 \times (P/A,\ i^*,\ 5)$$

$$(P/A,\ i^*,\ 5) = 80,000 / 27,400 = 2.9197$$

$$(P/A,\ 20\%,\ 5) = 2.9906; \quad (P/A,\ 22\%,\ 5) = 2.8636$$

$$AT\text{-}IRR = 20 + 2 \times (2.9906 - 2.9197)/(2.9906 - 2.8636) = 20 + 1.12 = 21.12\%$$

After-Tax IRR = 21.12%. If MARR_AT = 10%, the project is acceptable (21.12% > 10%).

Before-tax comparison: BT-IRR ≈ 34%. Taxes reduce IRR from 34% to 21% — a dramatic impact, demonstrating why after-tax analysis is essential.

Q14. Compare AT-NPW using SL and SYD. B = Rs. 60,000; S = 0; n = 4 years; BTCF = Rs. 28,000/yr; t = 35%; MARR = 10%. [2079 Chaitra Q10, 2077 Chaitra Q10 — 6 Marks]

Answer:

B = Rs. 60,000; S = 0; n = 4 yrs; BTCF = Rs. 28,000/yr; t = 35%; MARR = 10%

SL: d = 15,000/yr; SYD: SOYD = 10; d yr1-4: 24,000; 18,000; 12,000; 6,000

SL Method

TI = 13,000/yr; Tax = 4,550/yr; ATCF = Rs. 23,450/yr

AT-NPW_SL = -60,000 + 23,450(P/A,10%,4) = -60,000 + 23,450(3.1699) = Rs. 14,334

SYD Method

Year	BTCF (Rs.)	d_SYD (Rs.)	TI (Rs.)	Tax @35% (Rs.)	ATCF (Rs.)
1	28,000	24,000	4,000	1,400	26,600
2	28,000	18,000	10,000	3,500	24,500
3	28,000	12,000	16,000	5,600	22,400
4	28,000	6,000	22,000	7,700	20,300

AT-NPW_SYD = -60,000 + 26,600(0.9091)+24,500(0.8264)+22,400(0.7513)+20,300(0.6830)

= -60,000 + 24,182+20,247+16,829+13,865 = Rs. 15,123

Method	Total Tax (Rs.)	AT-NPW (Rs.)
SL	18,200	14,334
SYD	18,200	15,123 (better by Rs. 789)

SYD yields Rs. 789 higher AT-NPW through earlier, larger tax deductions. Total lifetime tax is identical — only the timing differs, and timing matters because of the time value of money.

Quick Reference — Chapter 7

Method	d_k Formula	BV at Year k
SL	(B−S)/n (constant)	B − k(B−S)/n
DB	(m/n)×BV_{k-1} (BV ≥ S)	B(1−m/n)^k
DDB (m=2)	(2/n)×BV_{k-1} (BV ≥ S)	B(1−2/n)^k
SYD	[(n−k+1)/SOYD]×(B−S)	B − cum. SYD
UOP	(B−S)/Q_total × q_k	B − cum. UOP

SOYD = n(n+1)/2

TI = BTCF − D_k | T = t×TI | ATCF = BTCF − T = NATI + D_k

Tax shield per year = t × d_k | AT-NPW = Σ [ATCF_k × (P/F, MARR_AT, k)]

- Accelerated methods (DDB, SYD) give same total tax as SL but earlier deductions → higher AT-NPW.
- BV may never fall below salvage value S.
- When actual salvage at disposal > BV, the excess is taxable recaptured depreciation.

—— End of Chapter 7: Depreciation and Income Tax ——

Tribhuvan University | Institute of Engineering | Engineering Economics (CE 655)